PROJECT OVERVIEW / THE HUMAN PROBLEM

# Polaris
# Ask the playbook.

A conversational way to find useful guidance inside a multi-page PDF.

### The problem

To answer a question, a reader opens the playbook, guesses search terms, scans several sections, and pieces the information together. Related questions can mean repeating the search.

### The solution

Polaris lets the user ask in plain language. It retrieves relevant passages from the playbook and uses them to help compose a focused answer.

### Tools used

n8n / Google Drive / Pinecone
OpenAI chat + embeddings / Simple Memory

**Portfolio summary**
I built a PDF knowledge assistant with n8n, Pinecone, and OpenAI that helps users find playbook guidance through natural-language questions and follow-ups.

SOURCE DOCUMENT




The Zero Lag VA Onramp Playbook is the source material. The supplied Google Drive preview shows a nine-page PDF.

STEP 01 / PREPARE THE PDF

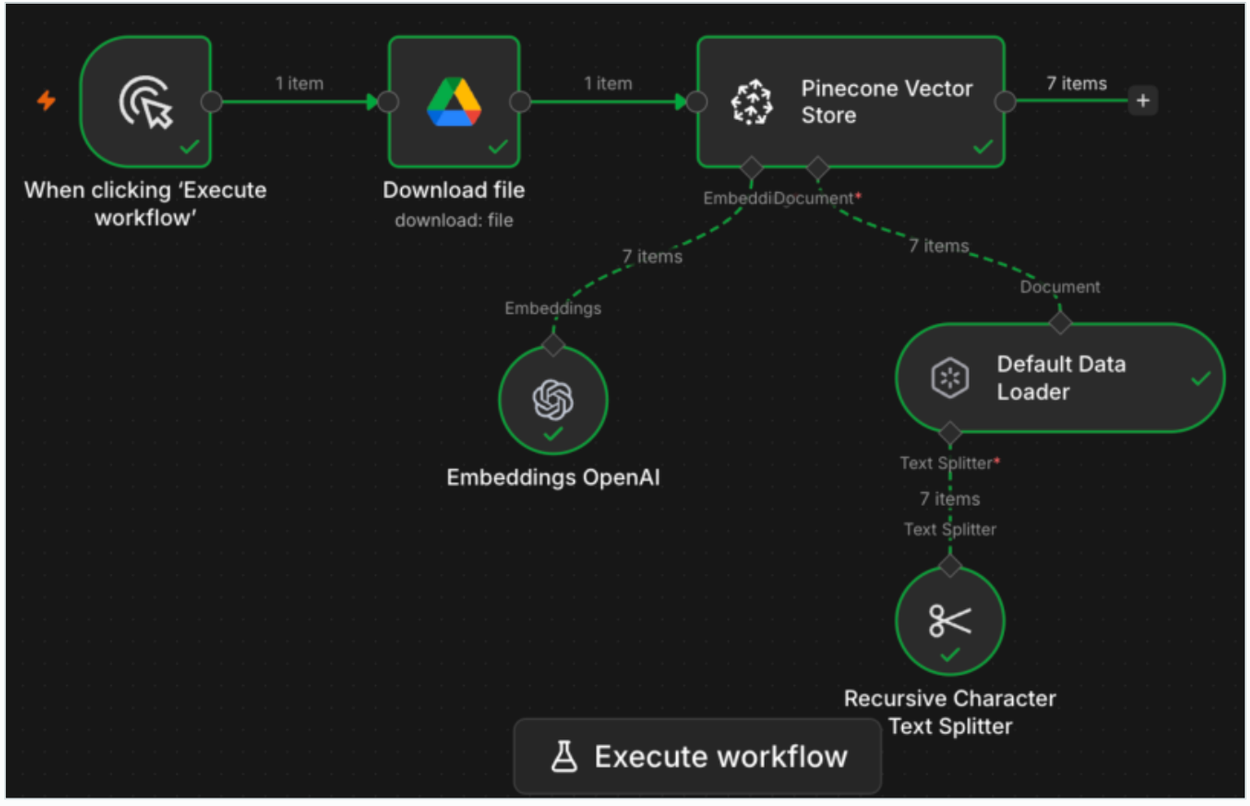
# Prepare the PDF once

The loading workflow turns the source PDF into searchable passages before users begin asking questions.

- **Start the workflow.** A manual trigger begins the setup run.
- **Download the file.** The Google Drive node retrieves the PDF.
- **Read and split.** The Default Data Loader reads the document; the Recursive Character Text Splitter divides the text into smaller passages.
- **Create embeddings.** OpenAI converts the passages into numerical representations used for meaning-based search.
- **Store the knowledge.** The Pinecone Vector Store receives the passages and their embeddings.

**Prepare once per document version**
Users do not repeat this setup for every question. Refresh the stored content when the playbook changes.

N8N / PINECONE VECTOR WORKFLOW




The loading workflow connects Google Drive, a data loader, a text splitter, OpenAI embeddings, and Pinecone.

RECORDED LOADING RUN




The supplied execution shows Success in 6.719s. Pinecone reports seven output items; this is evidence of a successful loading run.

STEP 01 CONTINUED / SEARCHABLE KNOWLEDGE

# Give the PDF a searchable memory

Pinecone holds the searchable knowledge. Think of it as a library index that helps the workflow locate relevant passages.

### What is an embedding?

An embedding is a numerical representation of text. The workflow uses these representations to look for passages related to the meaning of a question.

### Why split the PDF?

Smaller passages give the agent focused material to work with, instead of asking it to handle the entire file for every question.

### What the screenshot shows

Two Pinecone indexes are visible: **knowledge2** and **knowledge**. Their displayed dimensions are 1536 and 512, with **text-embedding-3-small** listed as the embedding model.

**Keep loading and search aligned**
Use the same target index and compatible embedding settings for both workflows. The index dimension must match the vectors being stored and searched.

PINECONE / INDEX CONFIGURATION

INDEX: KNOWLEDGE2




Displayed configuration: 1536 dimensions, on-demand capacity, AWS us-east-1, and text-embedding-3-small.

INDEX: KNOWLEDGE




Displayed configuration: 512 dimensions with the same model label. Host identifiers are hidden for portfolio presentation.

**Knowledge storage is different from chat memory.** Pinecone stores document knowledge; Simple Memory keeps conversational context for the chat session.

STEPS 02-03 / ASK AND RETRIEVE

# Ask naturally. Find relevant text.

### Step 2: Ask a question

A message such as "What should I do in the first week?" enters through the **When chat message received** trigger and reaches the Polaris Agent.

### Step 3: Find relevant passages

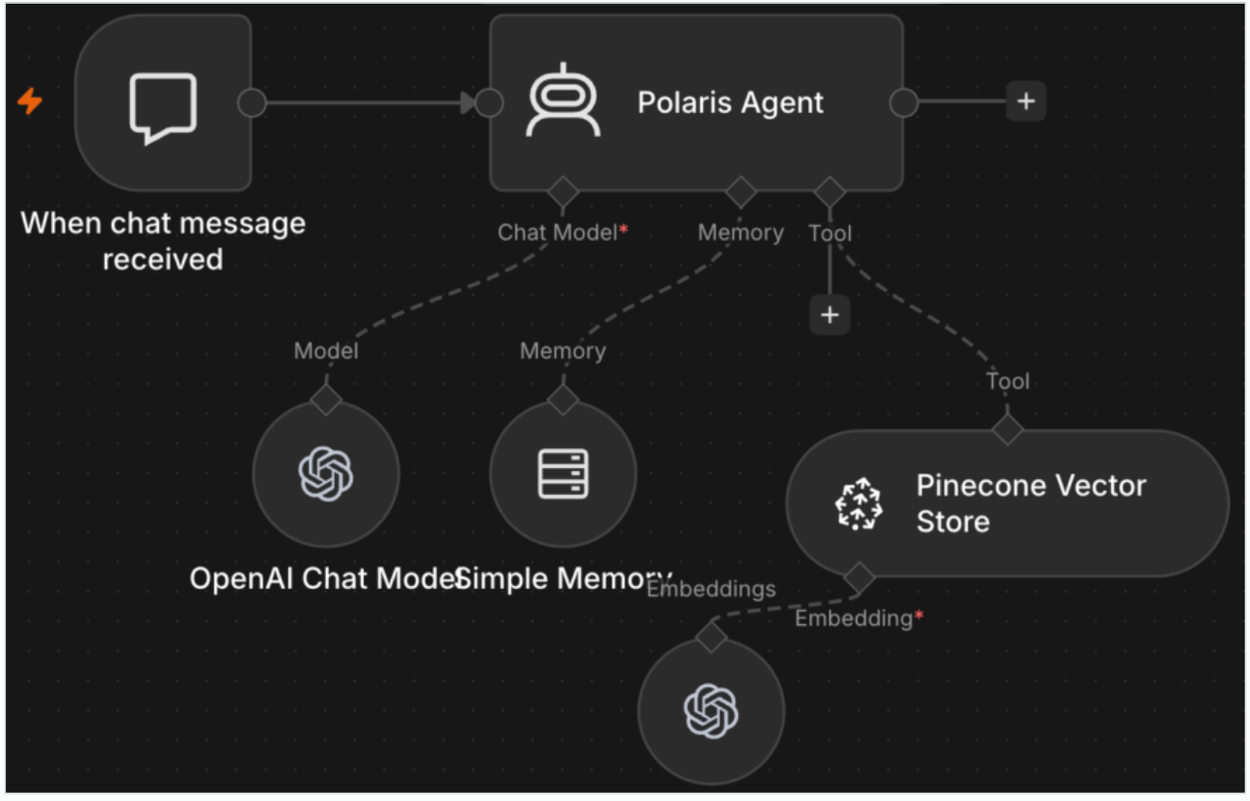
The agent can call Pinecone as a tool. The connected embeddings node supports comparing the question with the stored document passages.

### Search by meaning

The user does not have to guess the author's exact wording. The goal is to retrieve passages about the same topic, even when the question uses different words.

**RAG, in plain language**
Retrieval-augmented generation means: look up useful source text first, then use it to help write the answer.

N8N / POLARIS AGENT WORKFLOW




The chat workflow connects the agent to an OpenAI Chat Model, Simple Memory, and the Pinecone Vector Store. Pinecone also uses an embeddings node.

EXAMPLE QUESTIONS / ILLUSTRATIVE

"What should I do in the first week?"

"Which part of the playbook explains onboarding?"

"Can you help me understand that section?"

The screenshot shows the n8n editor and chat setup; public deployment is not demonstrated here.

STEPS 04-05 / ANSWER AND CONTINUE

# Explain clearly. Keep the context.

### Step 4: Write a useful answer

The OpenAI Chat Model helps the agent turn retrieved passages into a concise response. The walkthrough specifies GPT-4o as the chat model.

### Stay within the source

The intended behavior is to acknowledge when the retrieved material does not support an answer and avoid filling the gap with invented details. Grounding helps, but important claims still need source checks.

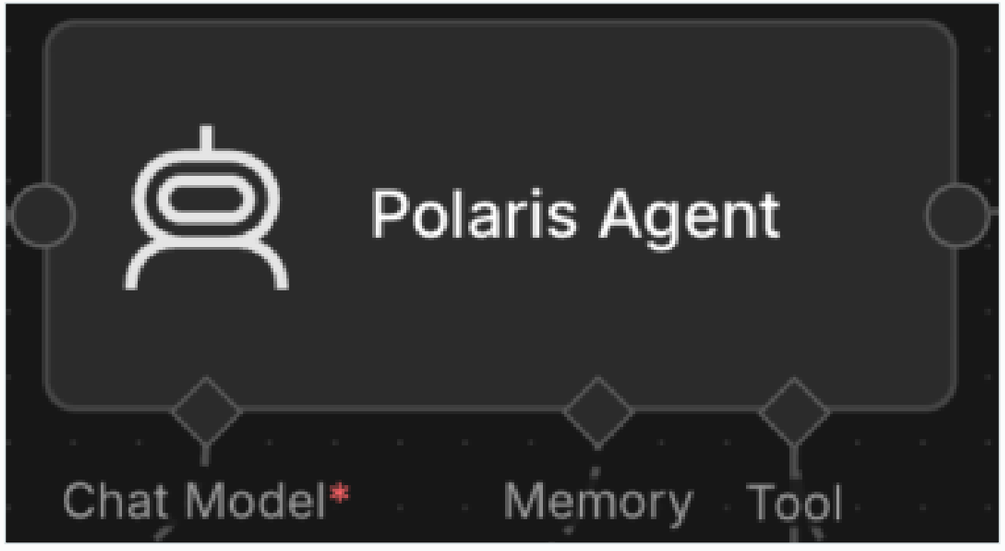
### Step 5: Ask a follow-up

Simple Memory supplies conversation context, so a related question such as "Can you turn that into a checklist?" can build on the preceding exchange.

> **Two kinds of context**
> Document passages supply the subject knowledge. Conversation memory helps interpret what "that" or "an example" refers to within the session.

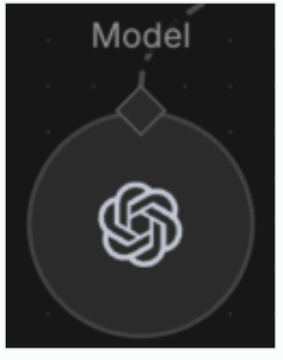
AGENT COMPONENTS / DETAIL VIEWS

POLARIS AGENT / MODEL, MEMORY AND TOOL INPUTS



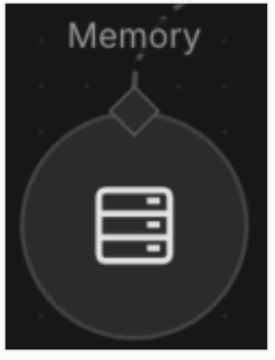

The agent connects the response model, conversation memory, and retrieval tool.

OPENAI CHAT MODEL




Compose the reply

SIMPLE MEMORY




Support related follow-ups

OUTCOME / BEFORE AND AFTER

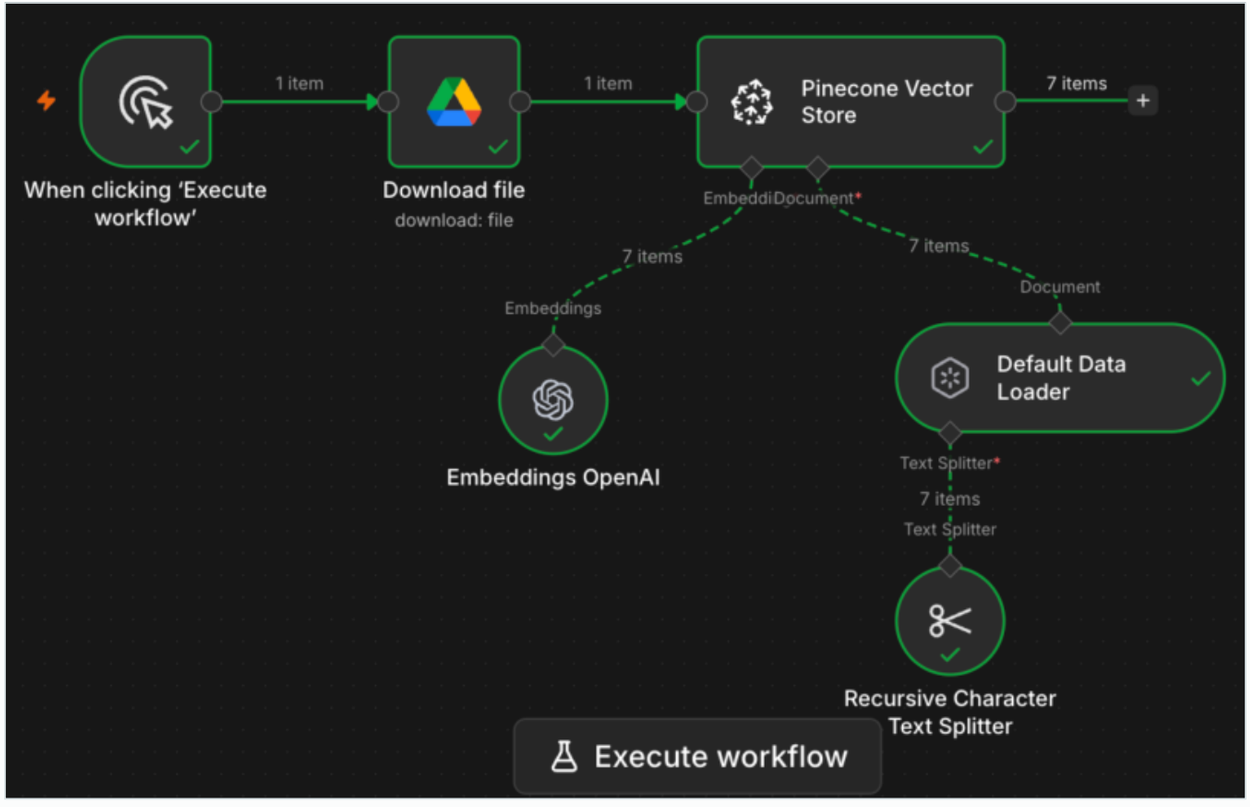
# Less searching. More useful guidance.

| TASK | BEFORE | WITH POLARIS |
|---|---|---|
| **Find** | Guess keywords; scan pages | Ask in plain language |
| **Understand** | Combine several sections | Read a focused reply |
| **Follow up** | Search the file again | Continue the session |
| **Verify** | Check the original text | Check the source behind the reply |

## USEFUL NEXT IMPROVEMENTS

- **Source references.** Return a page or section reference with each answer.
- **Refresh + recovery.** Update changed PDFs and alert on failed loading or chat runs.
- **Real chat examples.** Capture a supported question, an unsupported question, and a follow-up to verify the behavior.

## WHAT THE SCREENSHOTS DEMONSTRATE




The loading screenshot shows connected nodes with successful execution indicators and seven items at the Pinecone output.

## EVIDENCE AND EXPECTED BENEFITS

**Shown:** a source PDF, Pinecone indexes, a successful loading run, and a connected chat-agent architecture.

**Expected:** less repeated searching and easier follow-ups. Answer accuracy and time savings are not measured by these screenshots.